



Artificial Intelligence in Clinical Pharmacy Practice

Pharm.D Vault Editorial Team | Educational Review Draft

Abstract. Artificial intelligence (AI) is increasingly used in medication-related workflows, including order review, adverse-event detection, dose support, and prioritization of clinical alerts. This educational review describes where AI may assist clinical pharmacists, the safeguards needed before patient-facing use, and a practical framework for implementation. Current evidence supports cautious use of AI as a decision-support layer rather than as an autonomous prescriber. Clinical pharmacist oversight, local validation, transparent performance monitoring, privacy protection, and clear escalation pathways remain essential.

Keywords: artificial intelligence; clinical pharmacy; medication safety; decision support; pharmacotherapy

Introduction

Clinical pharmacy decisions often depend on large and changing data sets: medication histories, laboratory values, diagnoses, renal and hepatic function, microbiology results, and clinical notes. AI methods can process patterns across these inputs and may help pharmacists identify patients who need prompt review. The aim is not to replace professional judgement. Rather, the most credible role for AI is to reduce repetitive work, prioritize attention, and make evidence review more efficient while the pharmacist remains accountable for the final recommendation.

Applications in the Medication Use Process

AI-enabled tools can support medication-order review by identifying combinations of patient characteristics and medicines associated with a higher likelihood of harm. Natural-language processing can extract medication-related information from free-text notes. Predictive models may help identify patients at elevated risk of an adverse drug event, non-adherence, or an inappropriate dose. In pharmacy operations, AI may assist stock forecasting, dispensing verification, and triage of clinical alerts. The evidence base is heterogeneous; therefore, a reported model performance metric should not be assumed to transfer to a different hospital, patient population, or electronic record system.

Improving Clinical Decision Support

Traditional clinical decision support can generate many alerts of limited value. When alerts are frequent and poorly targeted, clinicians may become desensitized. AI-based prioritization may help identify alerts that deserve rapid review, but a high-priority label should be explainable. Pharmacists need to understand the data inputs, intended use, known limitations, and the action expected after an alert. A safe design displays the clinical rationale and allows the pharmacist to accept, modify, or reject the recommendation without unnecessary workflow burden.

Governance, Equity, and Privacy

AI systems can reproduce bias when development data do not represent local populations or when historical practice reflects unequal access to care. Before implementation, organizations should specify the intended population, evaluate performance across relevant subgroups, and define an acceptable error threshold. Data governance should address access controls, retention, cybersecurity, and the handling of identifiable health information. Patient-facing or device-like decision-support functions may also be subject to regulatory requirements. Procurement should include evidence of validation, version control, change management, and post-deployment monitoring.

A Practical Pharmacist Led Implementation Pathway

A pharmacy service can start with a narrow, measurable use case such as triaging medication alerts or identifying patients due for renal-dose review. The team should establish baseline workflow measures, test the tool in silent mode, compare its outputs with pharmacist review, and document false positives and false negatives. After a supervised pilot, the service can define response times, escalation processes, staff training, and regular audit intervals. Decisions about expansion should depend on demonstrated clinical usefulness, not novelty alone.

Conclusion

AI can strengthen clinical pharmacy practice when it is integrated as transparent, validated decision support. The key test is whether it helps pharmacists deliver safer, more timely, and more equitable medication care. Organizations should implement gradually, monitor continuously, and retain pharmacist oversight for every patient-specific therapeutic decision.

References

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